

HCI History and Today's Opportunities – What We Anticipated, What We Did Not

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ABSTRACT

In a time of rapidly-changing global digital immersion, what can the past tell us? More than you might think. Not everything remains relevant, but technology still collides with unchanging human nature, challenges still arise in collaborating across disciplines, we still must integrate UX, Design, and AI. We should understand how unanticipated consequences emerged. We often see positive potential but not negative consequences. HCI draws on computer science, human factors, information systems, and information science. Design is integral. We interact with AI. You are on a trajectory into the future. I identify significant elements that you can consider and address.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → **Historical people**; **History of hardware**; **History of software**.

KEYWORDS

HCI, Human Factors, Information Systems, Information Science, Design, AI, History, Future

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1 BENEFITS OF UNDERSTANDING HCI HISTORY AND ITS RELATIONSHIP TO CHALLENGES TODAY

Participants will learn how and why disciplines that contributed to HCI differ in their objectives and methods. They will get a deeper sense of how human nature can constrain technology uses. We will cover important consequences of computing that were not anticipated, and conclude with key opportunities and challenges arising as we face the future.

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It is said that those who do not learn history are doomed to repeat it. This is often not true in the rapidly changing technological world. Fifteen years ago, a new HCI rose from the ashes of an Internet bubble. The past several years have been disruptive. Nevertheless, we risk reinventing things that failed before. We can be surprised by consequences that we could have anticipated. When problems arise from technology limitations, success is possible as technology advances. When the issues are rooted in human cognitive and social behavior, history often will repeat itself. When organizational behavior was a key factor, new efforts might succeed as cloud computing and globalization change how we work and live.

Recent years have seen major changes in our interactions with devices and software. Many are positive, but there is growing uneasiness about effects of their uses and about the organizations that control the technologies. As billions of people increase their reliance on technologies that change rapidly, unintended consequences can have profound effects that are difficult to anticipate. They can spread rapidly and be difficult to steer. By looking back to find patterns in the past, sometimes we can better design for the future.

2 INTENDED AUDIENCE

Most people taking this course have been students, so I don't assume familiarity with HCI history, but I am confident that this will benefit any researcher or developer from any field who has a focus on technology design and use. I've recently realized the significance of some historical events that I had lived through but not appreciated, so I think the course could interest anyone interested in thinking about career directions or engagement with the major issues confronting us, since they all involve humans and digital technology. It can be useful to understand what the different disciplines engaged in HCI bring. Determine which might be relevant to your work and how to talk with people in those area, and which are less promising. See how to position your work, learn from the past, and identify considerations that deserve attention. I started as a practitioner. I was motivated to explore HCI history to understand things I learned the hard way. Some of those lessons are still useful. Everyone can profit from understanding the forces and tradeoffs that shape the HCI trajectories we are following.

3 PREREQUISITES

No prerequisites. There will be opportunity to comment and ask questions. Hands-on work comes afterwards, as you go forward to make new history.

4 CONTENT

This 75-minute lecture course (in-person or hybrid) will invite questions or comments prior to and during the course. The slides minimize bullet points, relying on timelines, images, and quotations. Part of the course follows the outline of my 2017 book, *From Tool to Partner: The Evolution of Human-Computer Interaction* [1]. Events over the past five years have not only revealed new opportunities and confronted us with new challenges, they have revealed the critical importance of past events that were not evident to me in 2017.

We start with the ongoing HCI challenges arising from changes in technology that can collide with human nature, a familiar observation but one that deserves a careful look. The course then surveys HCI history across fields that have been significantly engaged, but do not always communicate. The history of human-computer interaction falls into three eras: Prior to 1980, when computer processing was very expensive but some envisioned the future potential, from 1980 to 2005 when lower cost led to the rapid expansion of discretionary use and the realization of the early visions, and the fundamentally new era of HCI that presents opportunities and challenges not previously imaginable. Both transitions were disruptive. I also emphasize events in the mid-90s that set in motion the most significant changes in our field.

Among the disciplinary considerations are the relationship of AI and HCI, and within HCI the balance of quantitative and qualitative approaches. More recent developments include the relationship of interaction design and visual design, and the growing diversity of customer and user goals. A longstanding challenge is the role of digital technologies in supporting and disrupting communities. A topic moving front and center with surprising ferocity is the proliferation of unintended negative consequences of globe-spanning work, in the hands of both well-intentioned and increasingly powerful bad actors, with a large gray spectrum in between.

The historical review covers the fields of human factors, management information systems, computer science, and information science. It draws from articles and chapters in *Annals of the History of Computing*, *AI Magazine*, *CACM*, *IEEE Computer*, *ACM Interactions*, and several handbooks which informed the book. It includes my recent reassessment of that past.

I have been updating the slides since the CHI 2022 presentation and I share them freely.

5 INSTRUCTOR BACKGROUND

For several years I edited and sometimes wrote a column on HCI history for *Interactions* magazine. I have published many articles, handbook chapters, and in 2017 a monograph, *From tool to partner: The evolution of human-computer interaction* [1]. This course carefully considers what has happened, and what I have realized, since 2017. I have also actively participated in all of the fields mentioned above, although primarily CHI, starting with the first conference.

6 RESOURCES

The course is intended to point people in the direction of additional resources. Anyone wanting to take off on their own is welcome to look at the history publications on jonathangrudin.com, check out

HCI history columns in *Interactions* (2006-2013), or my 2017 book, or write me for more specific ideas at jgrudin@jonathangrudin.com.

7 ACCESSIBILITY

This will be a lecture course, so there are only the basic familiar access issues. If contacted by a participant with an issue, I'll do what I can to accommodate.

REFERENCES

- [1] Jonathan Grudin. 2017. *From Tool to Partner: The Evolution of Human-Computer Interaction* (1st ed.). Springer, Cham, Germany. <https://doi.org/10.1007/978-3-031-02218-0>